# EXPLORATORY DATA ANALYSIS PROJECT
# MOVIE RATING ANALYSIS

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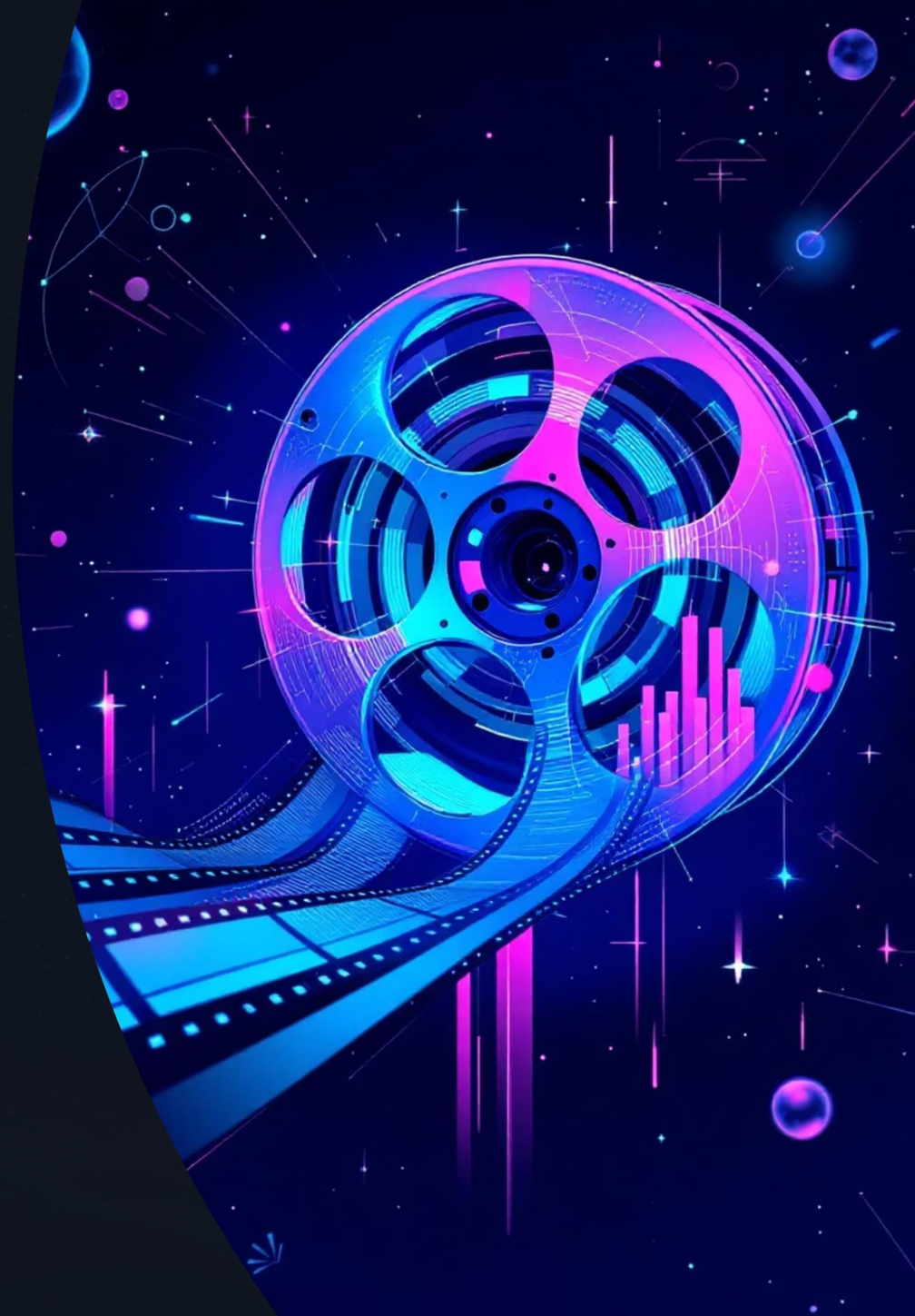
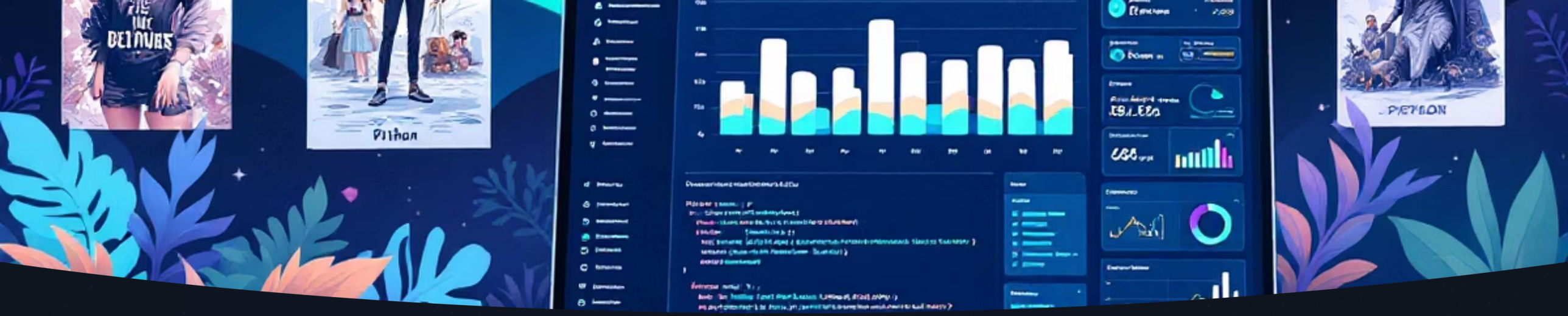
Internship Title: Data Science Intern
Tools Used: Python, Pandas,NumPy Matplotlib, Seaborn






# Introduction: Unveiling Movie Insights

This Exploratory Data Analysis (EDA) project aims to understand patterns in movie performance, audience ratings, industry trends, and financial outcomes.

### Purpose

Analyze movie performance, ratings, and financial success.

### Problem Statement

Identify key factors influencing movie success and audience reception.

### Tools & Libraries

Python, Pandas,NumPy Matplotlib, Seaborn.

# Dataset Overview: Movie Metrics

**The dataset contains movie-related features such as:**

**Rating**

**Budget**

**Run time**

**Writers**

**Genre**

**Year of release**

**Director**

| rank | name | year | rating | genre | certificate |
|---|---|---|---|---|---|
| 1 | e Shawshank Redemption | 1994 | 9.3 | Drama | R |
| 2 | The Godfather | 1972 | 9.2 | Crime,Drama | R |
| 3 | The Dark Knight | 2008 | 9.0 | Action,Crime,Drama | PG-13 |
| 4 | The Godfather Part II | 1974 | 9.0 | Crime,Drama | R |
| 5 | 12 Angry Men | 1957 | 9.0 | Crime,Drama | Approved |
| 6 | Schindler's List | 1993 | 9.0 | Biography,Drama,History | R |
| 7 | gs: The Return of the King | 2003 | 9.0 | Action,Adventure,Drama | PG-13 |
| 8 | Pulp Fiction | 1994 | 8.9 | Crime,Drama | R |
| 9 | he Fellowship of the Ring | 2001 | 8.8 | Action,Adventure,Drama | PG-13 |
| 10 | ood, the Bad and the Ugly | 1966 | 8.8 | Adventure,Western | Approved |
| 11 | Forrest Gump | 1994 | 8.8 | Drama,Romance | PG-13 |
| 12 | Fight Club | 1999 | 8.8 | Drama | R |
| 13 | e Rings: The Two Towers | 2002 | 8.8 | Action,Adventure,Drama | PG-13 |
| 14 | Inception | 2010 | 8.8 | Action,Adventure,Sci-Fi | PG-13 |
| 15 | The Empire Strikes Back | 1980 | 8.7 | Action,Adventure,Fantasy | PG |
| 16 | The Matrix | 1999 | 8.7 | Action,Sci-Fi | R |

**Data Cleaning :**

**Data was cleaned and pre-processed to ensure accuracy and consistency for analysis.**

# Univariate Analysis: Individual Feature Insights

### Rating Distribution

Most movies fall within a moderate rating range, indicating balanced audience perception.

### Genre Distribution

A few dominant genres show higher production frequency, reflecting industry preferences.

### Year-wise Releases

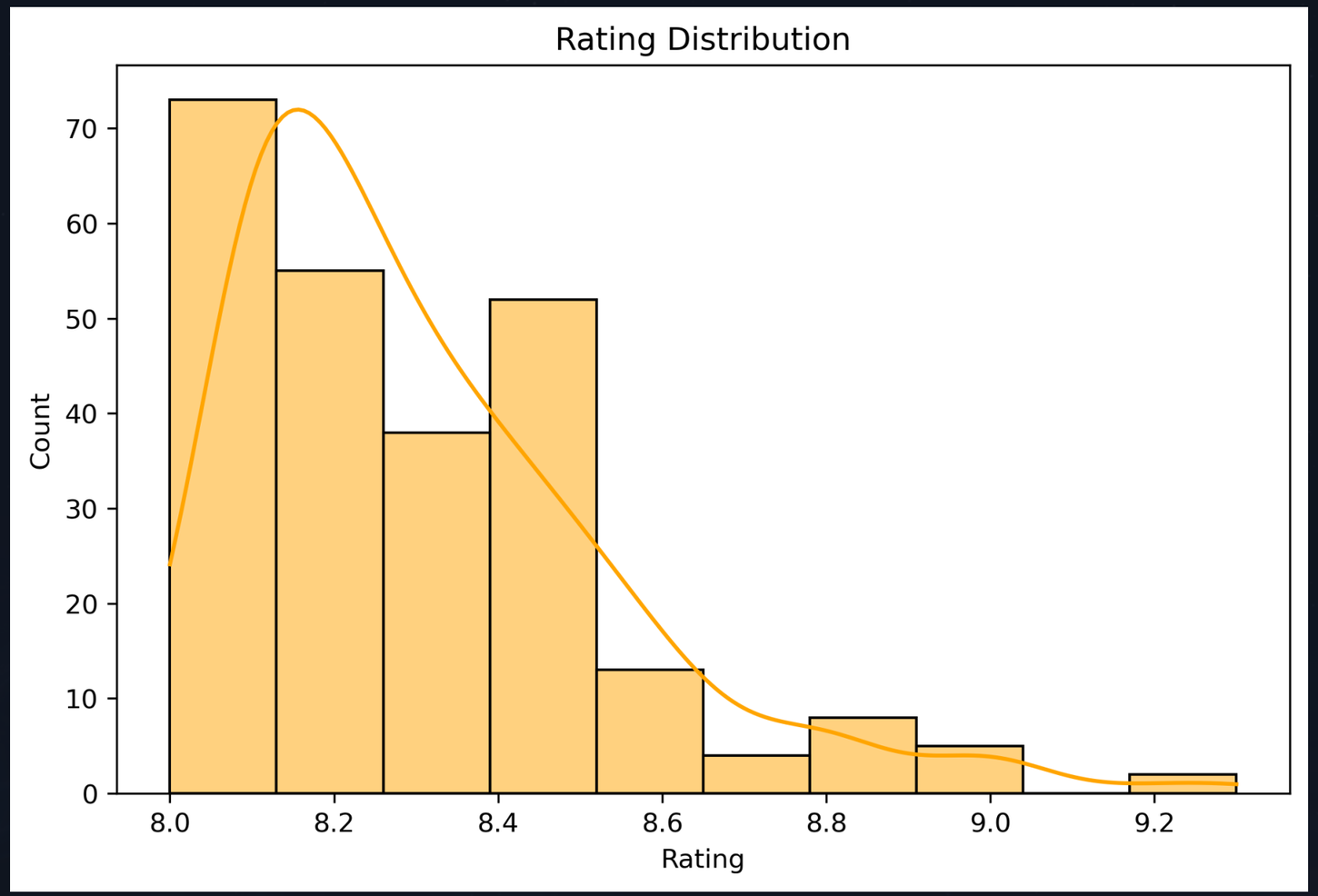
Rising trend over time, especially in recent decades, signifies industry growth.

### Top 10 Directors

Identifies prolific filmmakers and their significant influence on content creation.

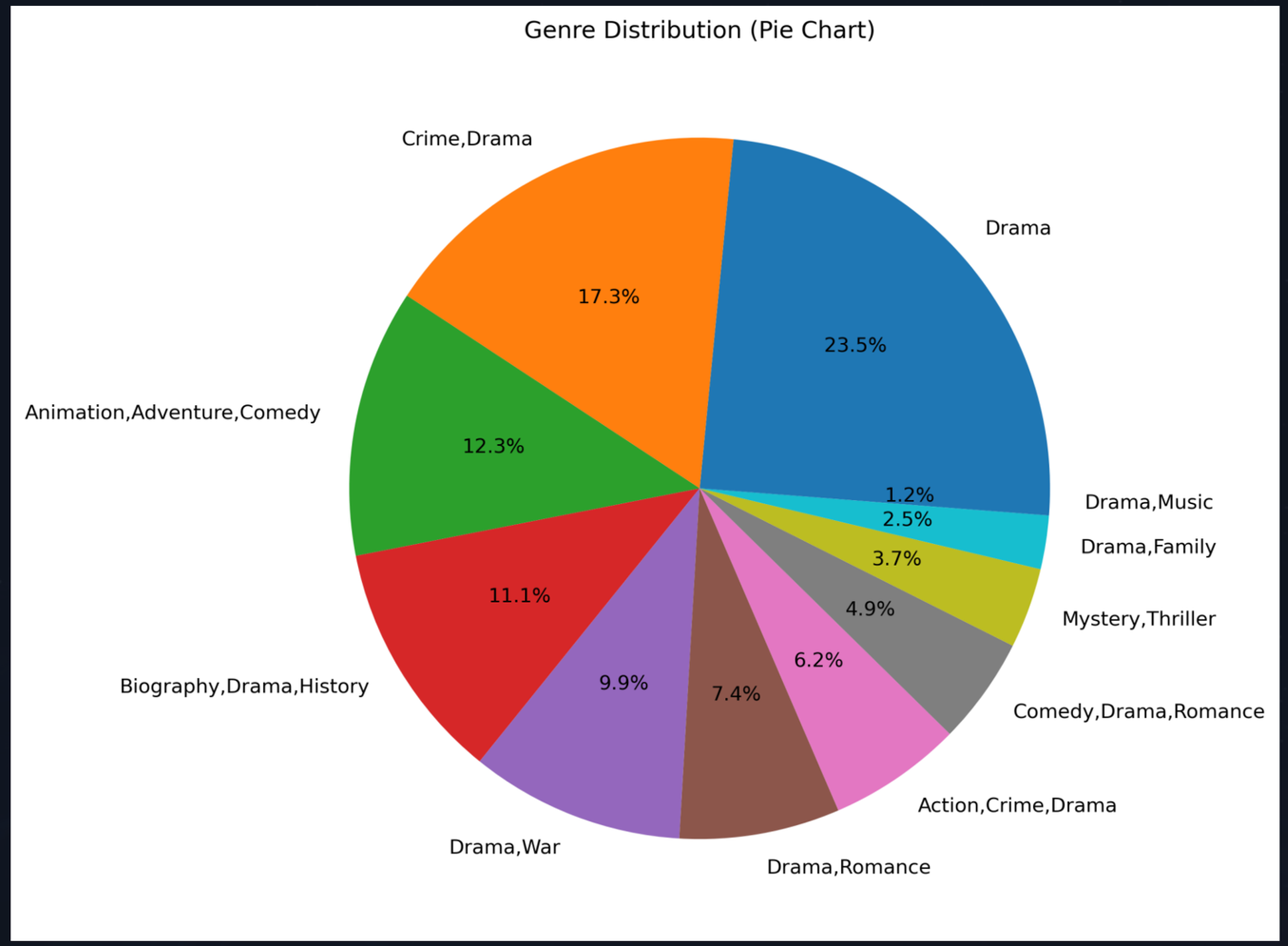
## RATING DISTRIBUTION:

- Most movies lie within a moderate rating range
- Very few movies receive extremely low or extremely high ratings .
- Indicates balanced audience perception towards most films



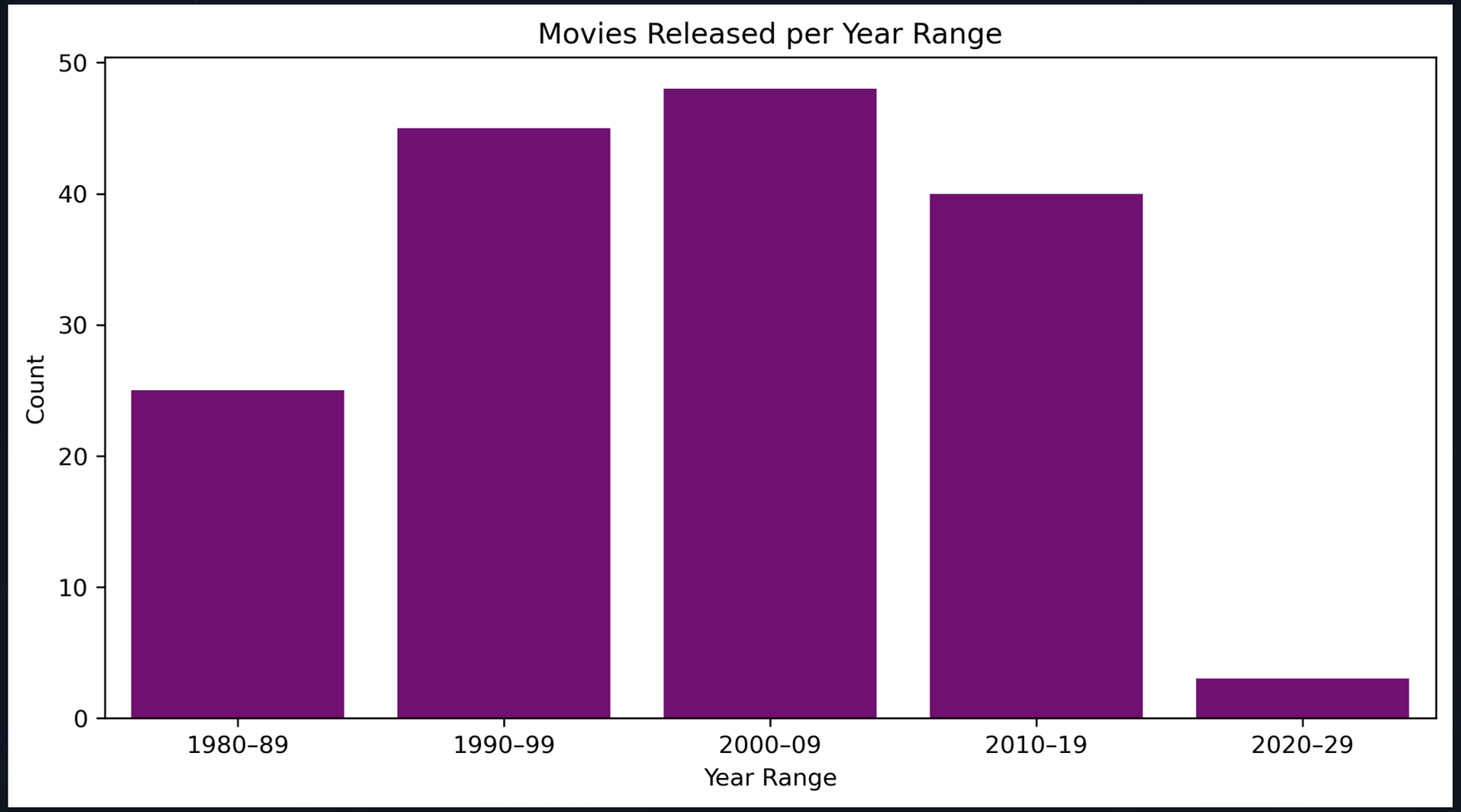
# GENRE DISTRIBUTION:

- A few genres dominate movie production.
- Indicates industry preference for commercially safe and audience-favorite genres.
- Shows which genres attract consistent investment



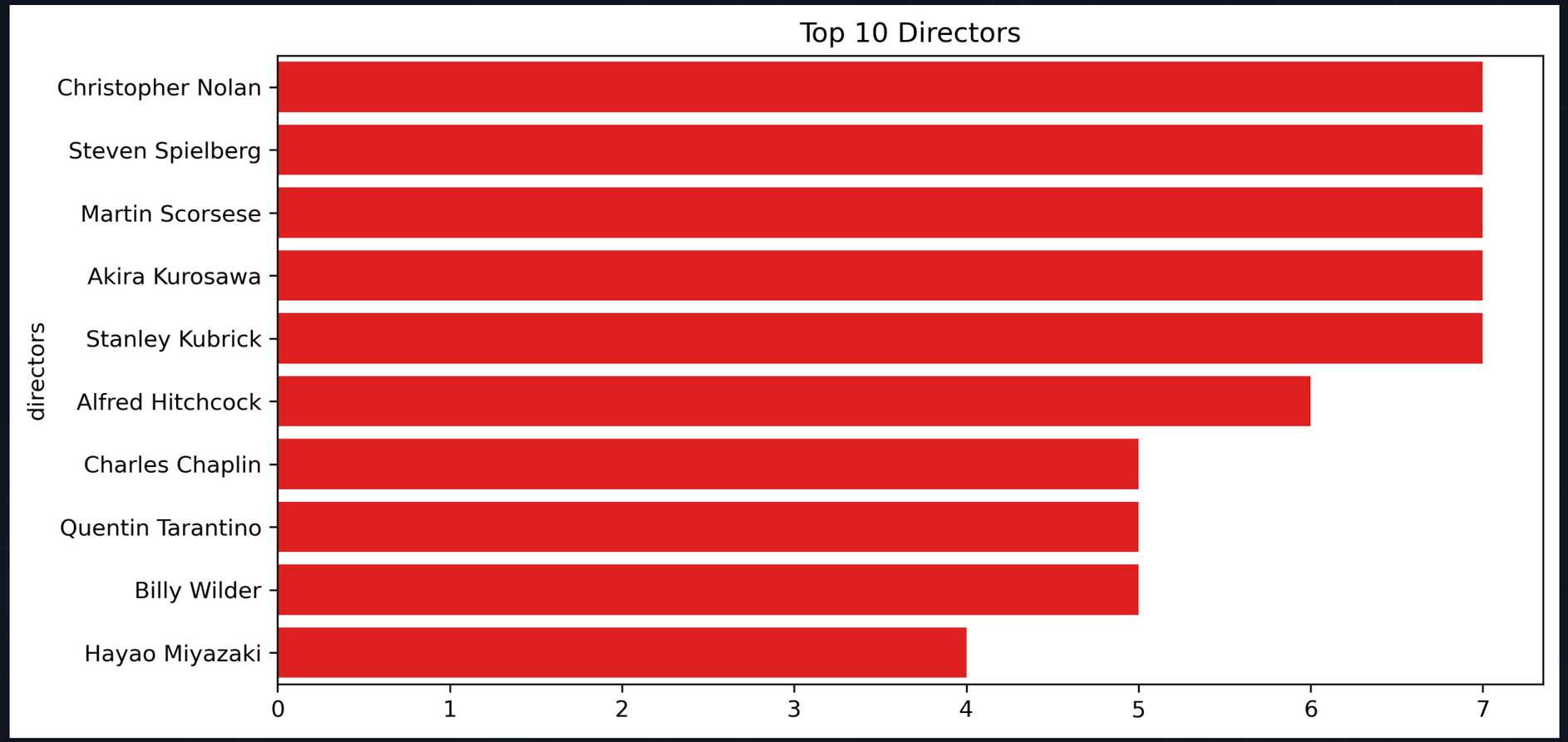
# YEAR WISE MOVIE COUNT:

- Movie production shows  a steady upward trend over the years.
- Modern decades indicate higher movie output due an to increased demand.
- Reflects expansion of the film industry



# TOP 10 DIRECTORS:

- Identifies filmmakers with the highest contribution to the dataset.
- Shows how a few prolific directors shape industry output.
- Demonstrates director influence on production patterns



# Bivariate Analysis: Interconnected Factors

## Budget vs. Box Office

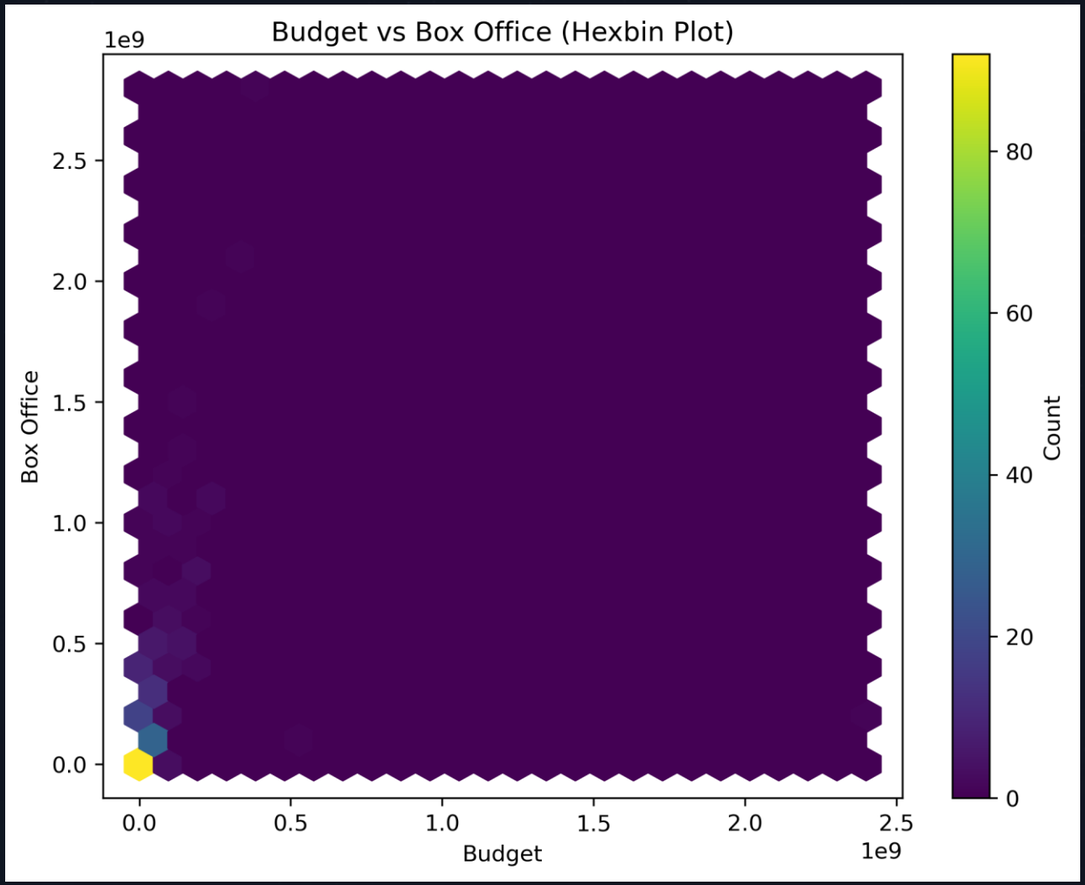
A clear positive relationship: higher budgets generally lead to stronger box-office earnings.

## Average Rating Year-wise

Average ratings remain consistent, with a significant upward trend observed between 1960-1980.

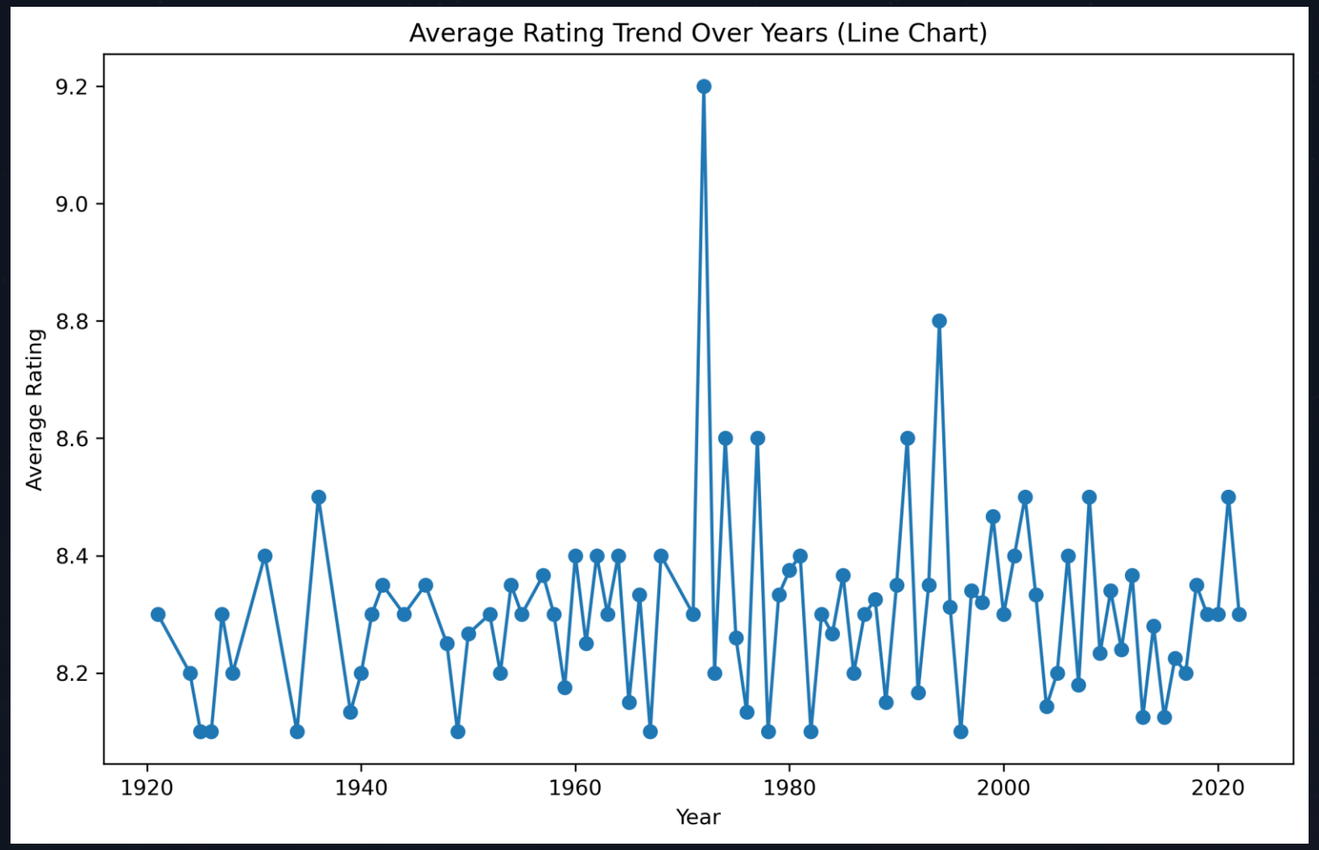
# BUDGET VS BOX OFFICE:

- A strong positive relationship is seen between budget and box office collections.
- Higher-budget films consistently generate higher revenue.
- Mid-to-high budget cluster is the densest → common investment zone.
- Confirms that investment is a major driver of commercial success.



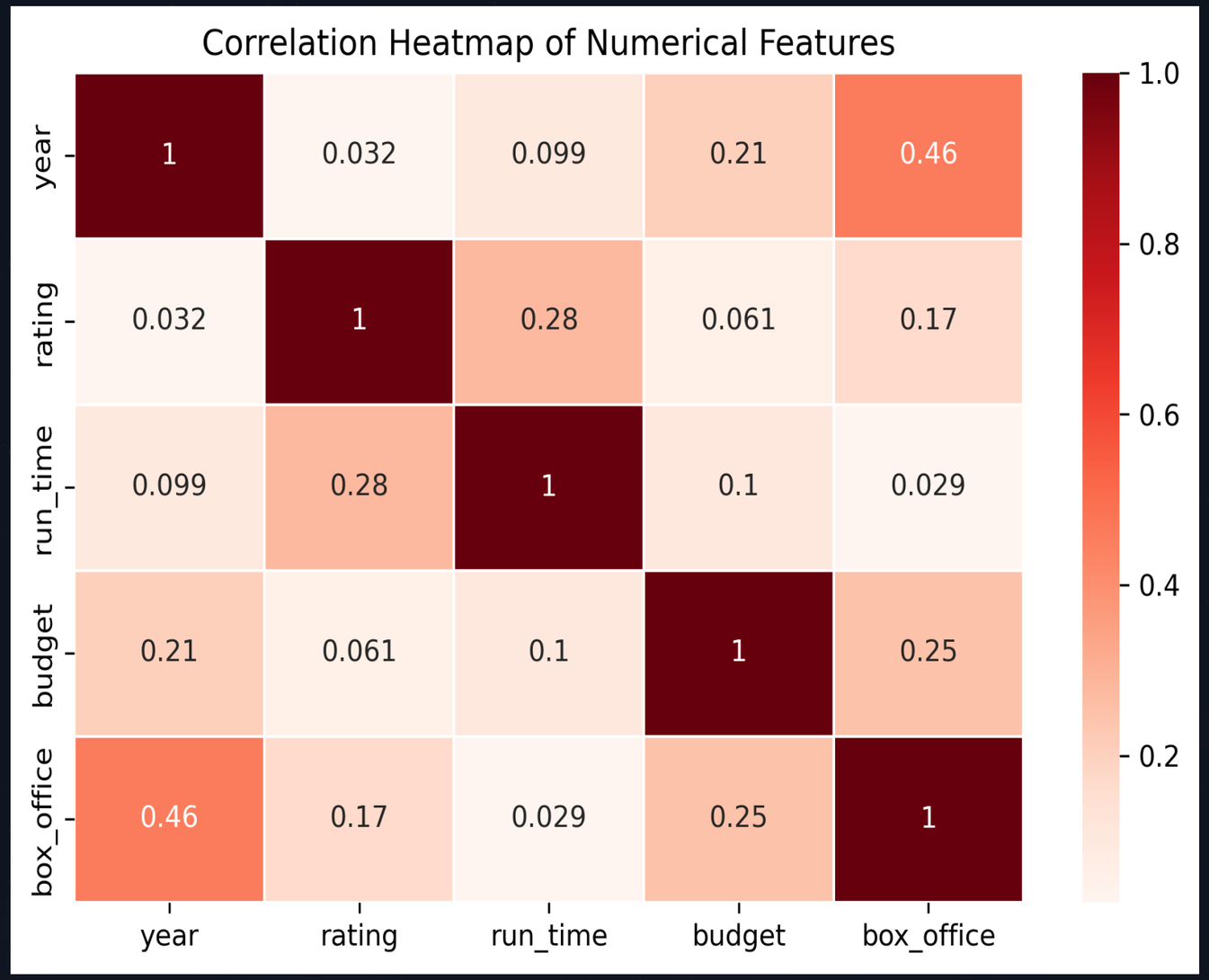
# AVERAGE RATING YEAR-WISE:

- Average ratings remain fairly stable across years.
- A noticeable rise occurs between 1960–1980, suggesting:
  - Strong storytelling
  - Unique themes
  - Indicates that audience standards have remained consistent.



# Multivariate Analysis: Correlation Heatmap

- Key observations:
  - Budget & Box Office: Strong positive correlation
  - Other numerical variables: Very weak correlation
- Runtime, rating, and year do not significantly
impact revenue.
- This highlights that financial success is investment-driven, not runtime or rating-driven



# Key Insights Summary

- **Ratings & Genres**
  Moderately distributed ratings; dominant genres reflect audience trends.
- **Industry Growth**
  Movie releases increased significantly, indicating industry expansion.
- **Director Influence**
  Top directors play a major role in shaping production volume.
- **Financial Success**
  Higher budgets drive higher box-office earnings; investment is key.
- **Audience Expectations**
  Average ratings remain stable, showing consistent audience reception.
- **Correlation**
  Only budget and box office show strong correlation; other relationships are weak.

## Conclusion

- The EDA reveals predictable trends in both creative and financial aspects of movie-making:
  - The industry continues to expand steadily.
  - Audience reception remains consistent across decades.
  - Budget emerges as the strongest predictor of box-office success.
  - Genre and experienced directors significantly influence production.

## Recommendations:

- Based on the analysis:
  - Invest more in genres with historically strong performance.
  - Provide higher budgets to projects aiming for strong commercial returns.
  - Collaborate with experienced and consistent directors.
  - Track year-wise trends to plan content aligned with audience expectations.